QISKIT FALL FEST 2026

FROM QUBITS TO PORTFOLIO DECISIONS

A three-level study of QAOA and quantum-kernel machine learning

TRACK 2 · QUANTUM OPTIMIZATION



TEAM

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MAP → OPTIMIZE → MEASURE

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PROJECT MAP

One problem, three learning levels

We increase the modeling complexity and change the quantum method to answer different questions.

01 BEGINNER

Can shallow QAOA find a known small solution?

6 stocks → choose 3

20 possible portfolios

METHOD

QAOA · p = 1

Simulator + IBM hardware

02 INTERMEDIATE

Can a quantum kernel represent portfolio similarity?

12 stocks → choose 6

924 candidate portfolios

METHOD

Fidelity kernel + SVM

Classical comparison + small hardware demo

03 ADVANCED

What changes when sector rules are enforced?

20 stocks → choose 10

7,776 feasible portfolios

METHOD

Constrained QAOA · p = 1

Exact baseline + simulator + hardware

PROGRESSION **Learn the algorithm → explore quantum similarity → stress-test constraints and hardware**

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HOW IT WORKS

From an investment question to a quantum experiment

The classical model defines the goal; the circuit samples candidate answers; classical code checks them.

CORE RISK–RETURN OBJECTIVE

$$C(x) = q \cdot x^T \Sigma x - \mu^T x$$

x: selected stocks · Σ: covariance (risk) · μ: expected returns · q: risk preference

ADVANCED CONSTRAINTS

Select exactly K stocks

$$\Sigma_i x_i = K$$

Sector cap

$$n_s \leq 2$$

Penalty terms discourage invalid selections.

THE EXPERIMENT PIPELINE

1. **Build data** — Synthetic returns + risk
2. **Encode cost** — QUBO / Hamiltonian
3. **Run circuit** — QAOA or feature map
4. **Measure** — Samples / kernel values
5. **Validate** — Compare to classical

All datasets in this project are synthetic; results are methodological demonstrations, not investment advice.

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LEVEL 1 · BEGINNER

A small search space makes validation exact

Six stocks, select three: enumerate all 20 candidates and use that optimum as the truth check.

CLASSICAL REFERENCE

Stock 2 · Stock 3 · Stock 6

Minimum objective cost

−0.503015

20 total portfolios · exact exhaustive search

VALID PORTFOLIO SAMPLES



Hardware optimum

29 / 1,024 shots

2.83% hit rate

Best valid hardware sample

Matched the exact optimum

QAOA depth p = 1 · IBM backend: ibm_kingston · 1,024 hardware shots

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LEVEL 2 · INTERMEDIATE

Quantum kernels measure portfolio similarity

This notebook explores machine learning, not QAOA: encode each portfolio, compare quantum states, then classify with an SVM.

QUANTUM-KERNEL PIPELINE

1. **Portfolio vector** — 12 binary stock choices
2. **Quantum feature map** — 12 qubits · 2 repetitions
3. **Fidelity kernel** — $K(x,y) = |\langle\varphi(x)|\varphi(y)\rangle|^2$
4. **SVM + ranking** — Compare with classical RBF SVM

924
candidate portfolios

100%
initial split · both models

1.4
quantum mean optimum rank

FIVE RANKING EVALUATIONS

Quantum kernel SVM	Mean Spearman 0.9469
Classical RBF SVM	Mean Spearman 0.9563

Classical baseline ranked the exact optimum first in all five recorded splits.

Exact reference optimum: Stocks 2, 6, 7, 9, 11, 12 · cost −1.066392.

Caution: 100% accuracy came from a synthetic, filtered-label task on the initial evaluated split; it is not evidence of real-world generalization.

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LEVEL 3 · ADVANCED

Scaling the model exposes the constraint problem

20 stocks, 10 selections, five sectors — with exhaustive search still feasible for this synthetic instance.

EXACT CLASSICAL BASELINE

7,776

feasible portfolios checked

Exact optimum holdings

S02 S04 S05 S07 S11

S12 S13 S16 S18 S20

Cost −1.111739

Return score 1.736840 · Risk 1.250202

QAOA SIMULATOR

10.30%	1 / 4,096	0.000000
valid samples	exact optimum hit	best-sample cost gap

Encoding + circuit facts

20 logical qubits · 211 Pauli terms · p = 1

Hamiltonian check: max error 5.116×10^{-13} on 101 bitstrings. Pilot search stopped at 20 evaluations; convergence was not established.

Sector rule: exactly two stocks from each of five sectors, implied by choosing 10 with a maximum of two per sector.

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REAL HARDWARE

Feasible sampling varied across our hardware experiments

These are different experiments, not a controlled apples-to-apples benchmark.

6 logical qubits

BEGINNER QAOA

59.47%

valid portfolio samples

609 / 1,024 shots

Exact optimum observed 29 times
Best valid sample matched classical optimum.

2-qubit demo

INTERMEDIATE QML

512

hardware shots

Feature-map-style circuit

Demonstrated circuit execution only;
full 12-qubit kernel/SVM was not run on hardware.

20 logical qubits

ADVANCED QAOA

0.39%

valid portfolio samples

1 / 256 shots

Exact optimum: 0 hits. Best valid cost −0.794773; gap 0.316966.

ADVANCED RUN: KEY LIMITATIONS

Depth 680 · 938 CZ gates · 256 shots · pilot angles not shown to have converged

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TAKEAWAYS

What we learned — and what we can honestly claim

The most important outcome is an end-to-end, classically checked workflow with real-device evidence.

THREE TAKEAWAYS

01 **A classical baseline is essential**

Exact enumeration tells us when a measured portfolio is truly optimal.

02 **Feasibility matters, not just cost**

Many measurements are invalid; decode and check constraints before comparing.

03 **Hardware results need context**

Circuit depth, shots, noise, and parameter quality shape what we observe.

SCIENTIFIC BOUNDARY

We demonstrated

QAOA formulation · simulator studies · IBM hardware runs · quantum-kernel SVM comparison

We did not establish

Quantum advantage, better-than-classical performance, or real-market investment value.

SELECTED REFERENCES **IBM Quantum: QAOA tutorial · Qiskit Finance: Portfolio Optimization · Qiskit ML: Quantum Kernel tutorial**

Havlíček et al. (2019), “Supervised learning with quantum-enhanced feature spaces,” Nature 567, 209–212. Full sources and experiment IDs: REFERENCES.md.

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